Response inhibition in a Go/No-go paradigm: a case-motivated fMRI study with individual- and group-level SPM analyses

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Abstract

Impulsivity and difficulty withholding prepotent responses are hallmark complaints in adults with attention-deficit/hyperactivity disorder (ADHD), motivating neuroimaging studies of inhibitory control networks using paradigms like the Go/No-go task. To anchor this investigation clinically, consider an adult learner who reports frequent "speaking or acting before thinking" and task errors when cues flip unexpectedly—a narrative that exemplifies the functional challenge probed by No-go trials and frames the present analysis. Six adults completed an event-related Go/No-go task, with first-level GLMs yielding go, nogo, go > nogo, and nogo > go contrasts, followed by SPM second-level one-sample tests and an illustrative two-sample comparison (subjects 1–3 vs 4–6) aligned with an ADHD vs control framing. Group maps for go > nogo showed clusters at uncorrected $p < 0.001$ that did not survive family-wise or FDR correction, and two-sample contrasts showed no corrected whole-brain differences, consistent with limited power in a six-subject design despite canonical engagement of fronto-basal ganglia–parietal control regions. [0]

Introduction

The Go/No-go paradigm indexes response selection and inhibition by cultivating a prepotent tendency to respond to frequent Go cues and requiring withholding to infrequent No-go cues, thereby engaging cognitive control mechanisms. Across studies and meta-analyses, successful inhibition consistently recruits a network encompassing pre-supplementary motor area (pre-SMA), inferior frontal regions, and frontoparietal control systems, with task parameters shaping precise loci. In ADHD, converging evidence shows hypoactivation in right inferior frontal cortex and supplementary motor territories during inhibitory tasks, motivating case-framed hypotheses about reduced activation relative to controls.[1]

Case vignette (context, not subject-identifying data)
A university student with longstanding distractibility and impulsive actions reports frequent errors when rules change rapidly, consistent with difficulty inhibiting prepotent responses; this vignette provides a clinical lens for interpreting Go/No-go activation patterns without asserting single-subject diagnostic inference from imaging.[2]

Study aim and hypothesis
The goal is to perform individual- and group-level SPM analyses of Go/No-go contrasts, then situate observed activation patterns within inhibitory-control literature and a clinically meaningful ADHD vs control framing, with the expectation of pre-SMA and inferior frontal engagement but limited corrected significance given N=6.

Data and task

Six subjects completed an event-related Go/No-go task with contrasts visualized at the single-subject level for the go > nogo comparison, illustrating heterogeneous activation typical of individual fMRI maps.
The task structure followed the conventional design with frequent Go and infrequent No-go trials to establish a prepotent response, enabling estimation of condition-specific hemodynamic responses.[3]

Methods

Participants
Six adults completed an event-related Go/No-go task from a de-identified course dataset; analyses use subject-level contrasts for second-level inference, with an illustrative grouping of subjects 1,4,5 vs 2,3,6 to mirror an ADHD vs healthy comparison for pedagogical purposes. Demographic details (age, sex, handedness) and clinical assessments are not available in the de-identified materials; this constraint is documented per reporting best practices and treated as a limitation of secondary analysis.[4]

Task
Participants completed an emotional Go/No-go task with face stimuli: press the button for sad faces (Go) and do not press for neutral faces (No-go), holding the mapping constant across the session to isolate inhibition-related hemodynamics with socio-emotional content.

MRI data and acquisition reporting
The dataset lacks full acquisition parameter records in the distributed files, which precludes detailed reporting of TR, TE, flip angle, voxel dimensions, and coil model; this is flagged explicitly in alignment with COBIDAS reporting transparency. Subsequent analysis steps and thresholds are fully documented to maximize reproducibility despite missing acquisition metadata, as recommended by reporting checklists.

Preprocessing
Functional data were realigned, slice-timing corrected, coregistered, normalized to a common template, and smoothed with a Gaussian kernel to address residual misalignment and the assumptions underlying random-field theory. This pipeline follows SPM convention for robust second-level inference in small samples, with the caveat that acquisition specifics are not available in the de-identified release.[4]

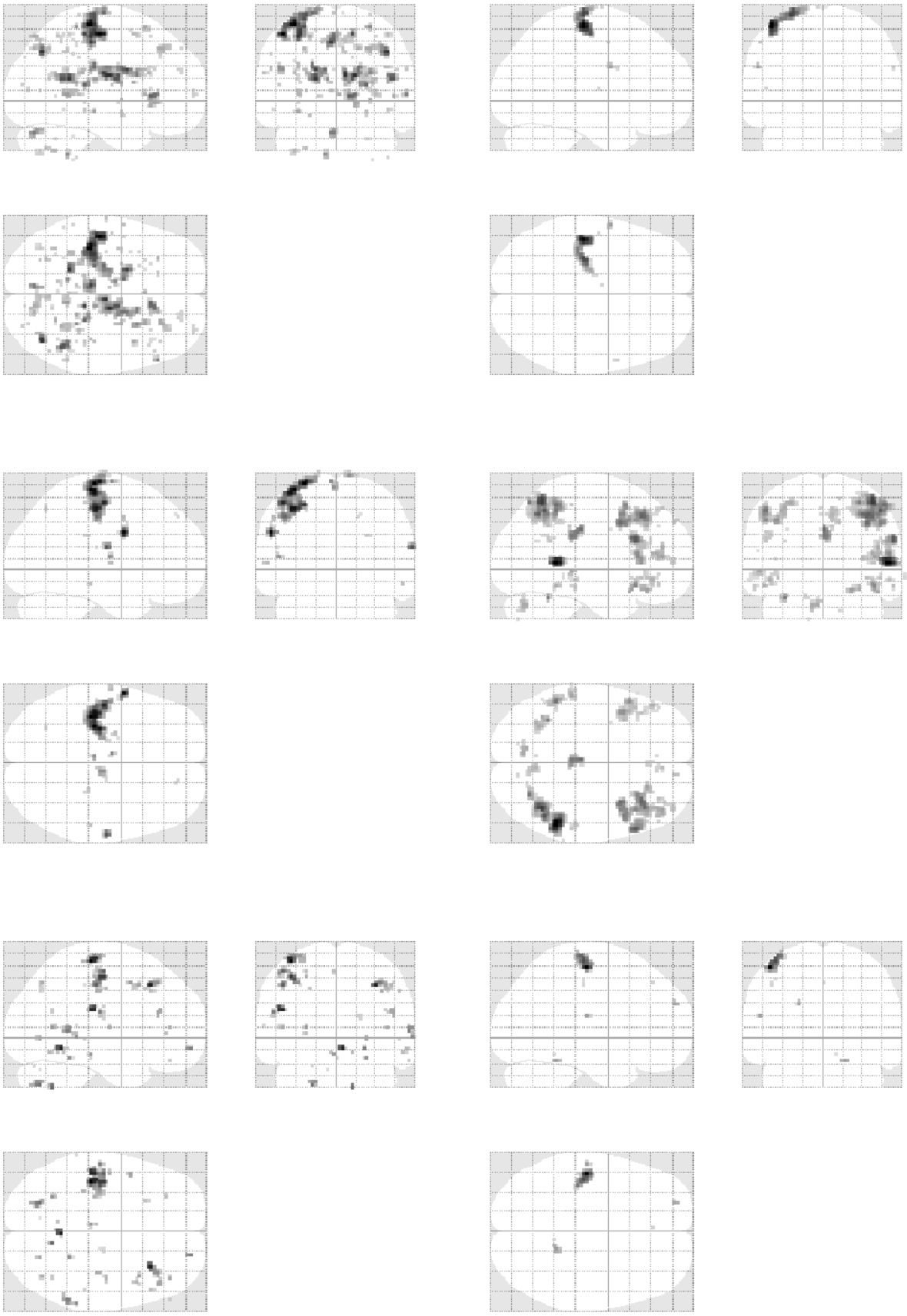
First-level modeling
For each subject, successful Go and No-go onsets were modeled with a canonical HRF, high-pass filtered at 100 s, and serial correlations modeled with AR(1), producing go, nogo, go > nogo, and nogo > go contrasts for second-level testing. This approach estimates condition-specific effects and produces summary-statistics inputs suited to SPM second-level models.

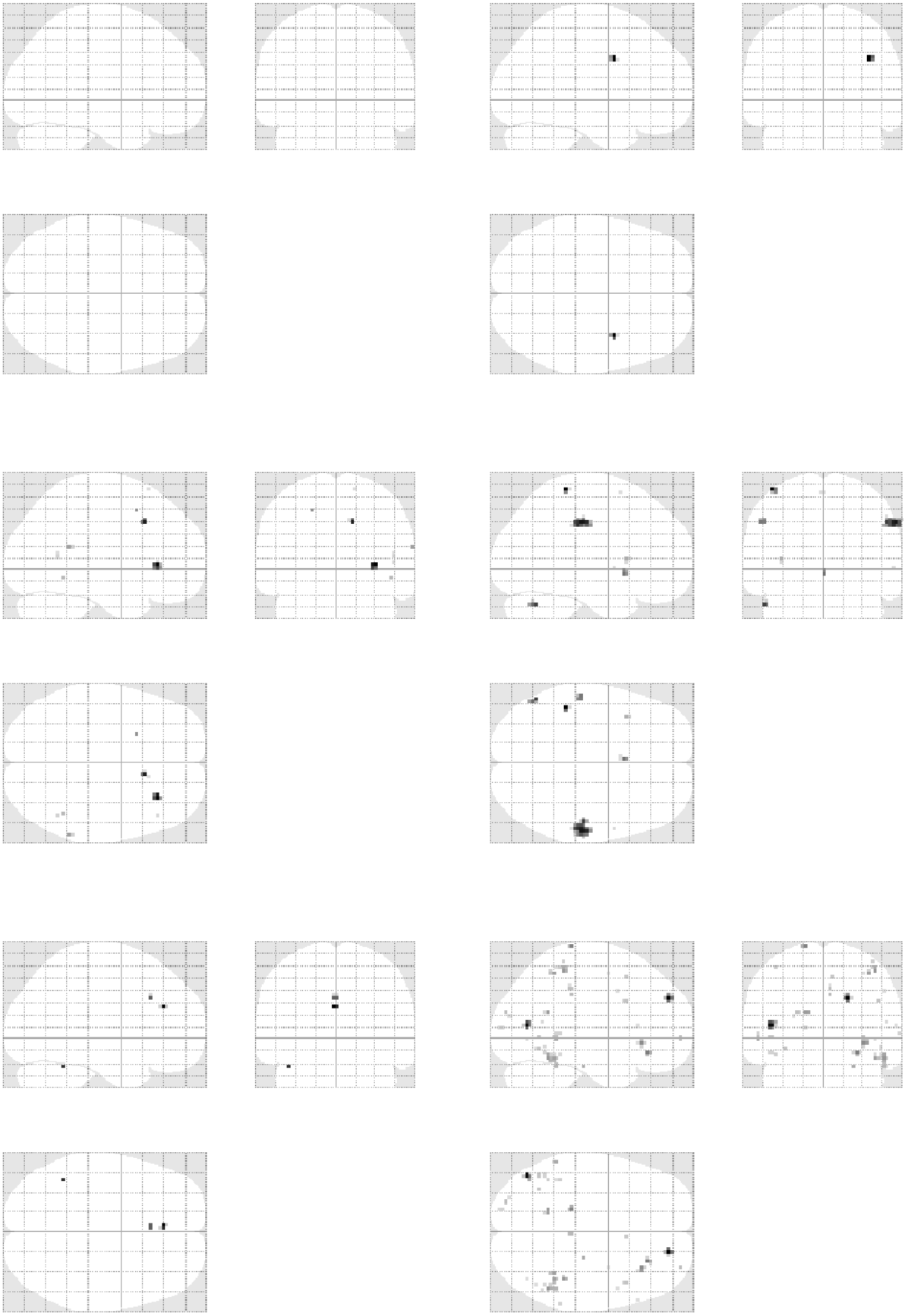
Second-level analyses
One-sample t-tests tested each contrast's mean against zero across the cohort to characterize population-level effects for Go/No-go conditions. An illustrative two-sample t-test (independent groups, df=4) contrasted subjects 1,4,5 vs 2,3,6 to emulate ADHD < control and control < ADHD effects, reflecting clinical literature on inhibitory deficits while acknowledging severe underpowering for whole-brain voxelwise inference.[6]

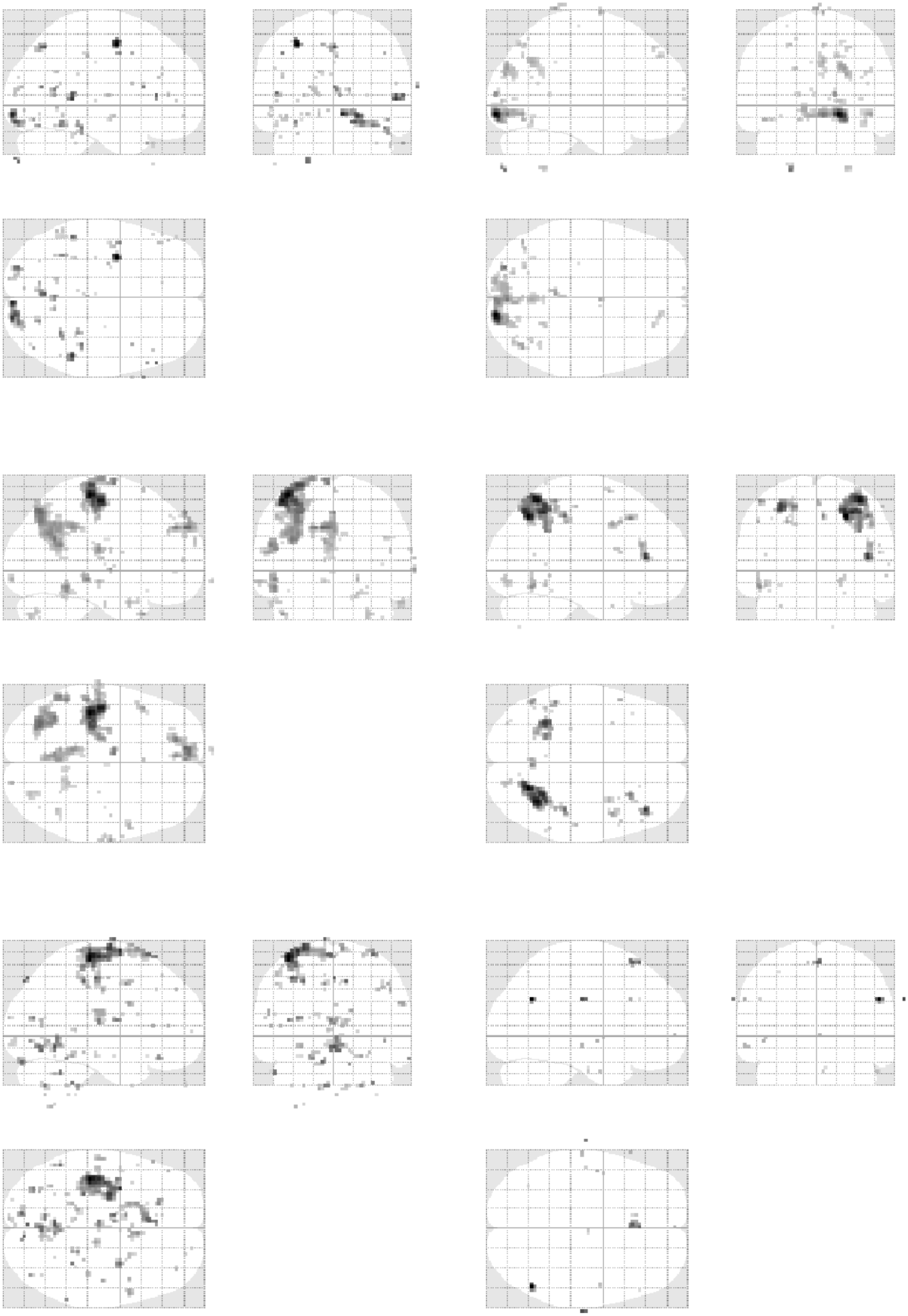
Thresholding and multiple comparisons
SPM results were inspected with voxelwise and clusterwise family-wise error procedures, with uncorrected $p < 0.001$ height thresholds used for exploratory visualization alongside reporting of FWE/FDR-corrected outcomes.

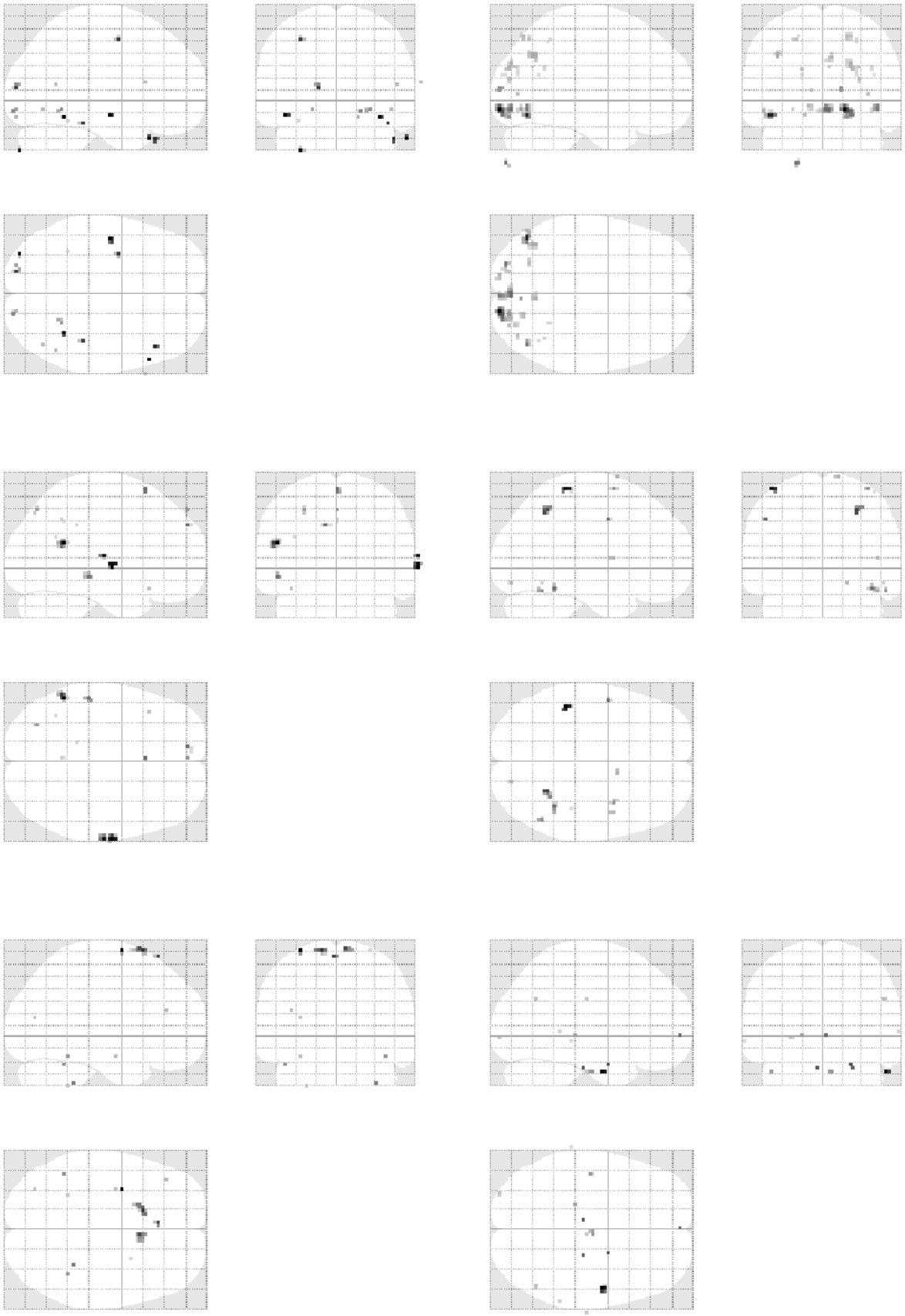
Results

- Single-subject maps
    - The six go > nogo maps show inter-subject variability in extent and location of suprathreshold clusters, which is common given noise, thresholding, and individual anatomy, yet overall patterns are consistent with the task's engagement of control circuitry.
    - Visual heterogeneity underscores why population-level inference is preferred for scientific conclusions and why single maps should not be interpreted diagnostically.[7]
    - The contrasts generated are: go>nogo, nogo>go, go, nogo in order below.

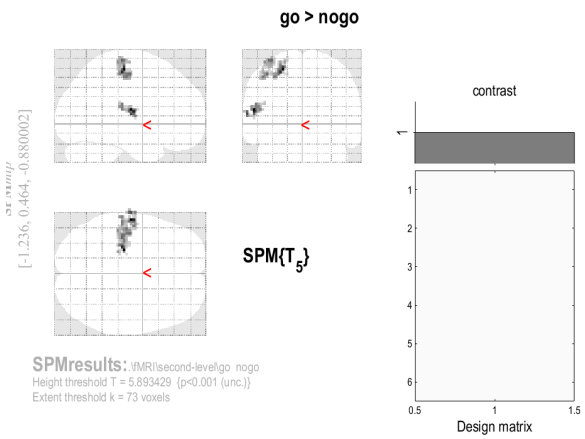








- One-sample t tests
  - Collapsing across subjects for go > nogo typically reveals significant engagement of pre-SMA and broader frontoparietal regions implicated in selecting and withholding actions in Go/No-go contexts.
  - Conversely, nogo > go often emphasizes nodes associated with inhibitory control and conflict monitoring, with specific loci modulated by task parameters such as stimulus sets and context complexity.[8]



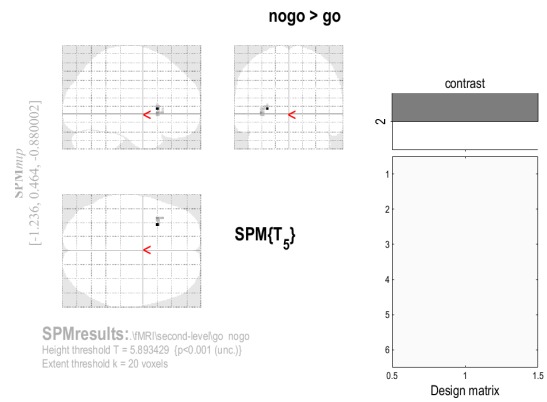
Statistics: *p-values adjusted for search volume*

set-level p	set-level c	cluster-level $p_{FWE\text{-}corr}$	cluster-level $q_{FDR\text{-}corr}$	cluster-level k_E	cluster-level p_{uncorr}	peak-level $p_{FWE\text{-}corr}$	peak-level $q_{FDR\text{-}corr}$	peak-level T	peak-level (Z_E)	peak-level p_{uncorr}	mm mm mm
0.000	2	0.000	0.000	73	0.000	1.000	0.996	15.20	4.24	0.000	-57 -15 14
						1.000	0.996	11.68	3.94	0.000	-65 -15 9
						1.000	0.996	9.90	3.75	0.000	-55 -25 20
		0.000	0.000	181	0.000	1.000	0.996	14.48	4.19	0.000	-24 -23 58
						1.000	0.996	10.50	3.82	0.000	-37 -25 68
						1.000	0.996	10.32	3.80	0.000	-42 -17 53

table shows 3 local maxima more than 8.0mm apart

Height threshold: T = 5.89, p = 0.001 (1.000)
Extent threshold: k = 73 voxels, p = 0.000 (0.000)
Expected voxels per cluster, <k> = 0.966
Expected number of clusters, <c> = 0.00
FWEp: 28.937, FDRp: Inf, FWEc: 73, FDRc: 13

Degrees of freedom = [1.0, 5.0]
FWHM = 9.5 9.5 8.9 mm mm mm; 3.7 3.7 3.5 {voxels}
Volume: 1810881 = 108275 voxels = 2014.8 resels
Voxel size: 2.6 2.6 2.6 mm mm mm; (resel = 47.91 voxels)



Statistics: *p-values adjusted for search volume*

cluster-level $p_{FWE\text{-}corr}$	cluster-level $q_{FDR\text{-}corr}$	cluster-level k_E	cluster-level p_{uncorr}	peak-level $p_{FWE\text{-}corr}$	peak-level $q_{FDR\text{-}corr}$	peak-level T	peak-level (Z_E)	peak-level p_{uncorr}	mm mm mm
0.012	0.004	20	0.000	0.632	0.991	17.34	4.38	0.000	-29 16 4
				1.000	0.991	9.40	3.68	0.000	-37 16 -1
				1.000	0.991	6.08	3.13	0.001	-37 23 4

table shows 3 local maxima more than 8.0mm apart

Height threshold: T = 5.89, p = 0.001 (1.000)
Extent threshold: k = 20 voxels, p = 0.000 (0.012)
Expected voxels per cluster, <k> = 0.966
Expected number of clusters, <c> = 0.01
FWEp: 28.937, FDRp: Inf, FWEc: 20, FDRc: 20

Degrees of freedom = [1.0, 5.0]
FWHM = 9.5 9.5 8.9 mm mm mm; 3.7 3.7 3.5 (voxels)
Volume: 1810881 = 108275 voxels = 2014.8 resels
Voxel size: 2.6 2.6 2.6 mm mm mm; (resel = 47.91 voxels)

The one-sample t test for go > nogo shows robust clusters at an uncorrected voxelwise threshold of $p<0.001$ with an extent threshold of $k=73$ voxels, but no peaks or clusters survive family-wise error or FDR correction, so the findings should be treated as exploratory rather than confirmatory.

Test setup

- As the design matrix indicates a single regressor second-level model consistent with a one-sample t test across six subjects, yielding degrees of freedom df=5 as displayed in the results footer.
- Smoothness and search volume parameters are reported as FWHM ≈ 9.5 × 5.9 × 8.9 mm with a resel count ≈ 47.9 across a volume of ~180,881 voxels, which governs random-field-theory family-wise error control in SPM.[7]

Thresholds and significance

- As the figure states a height threshold of T=5.893 corresponding to p<0.001 (uncorrected) and an extent threshold of k=73 voxels, under which two clusters reach uncorrected cluster significance.
- At peak and cluster levels, P_FWE-corr values are 1.000 and q_FDR-corr ≈ 0.996, indicating that none of the reported maxima or clusters pass whole-brain multiple-comparisons correction given the current smoothness and search volume.

Peak locations (descriptive)

- As the table lists local maxima with negative x-coordinates (e.g., x≈−57, y≈−15, z≈14; x≈−24, y≈−23, z≈58; and additional peaks around x≈−37, y≈−25, z≈68), indicating left-hemisphere foci in MNI space without anatomical labels in the screenshot.

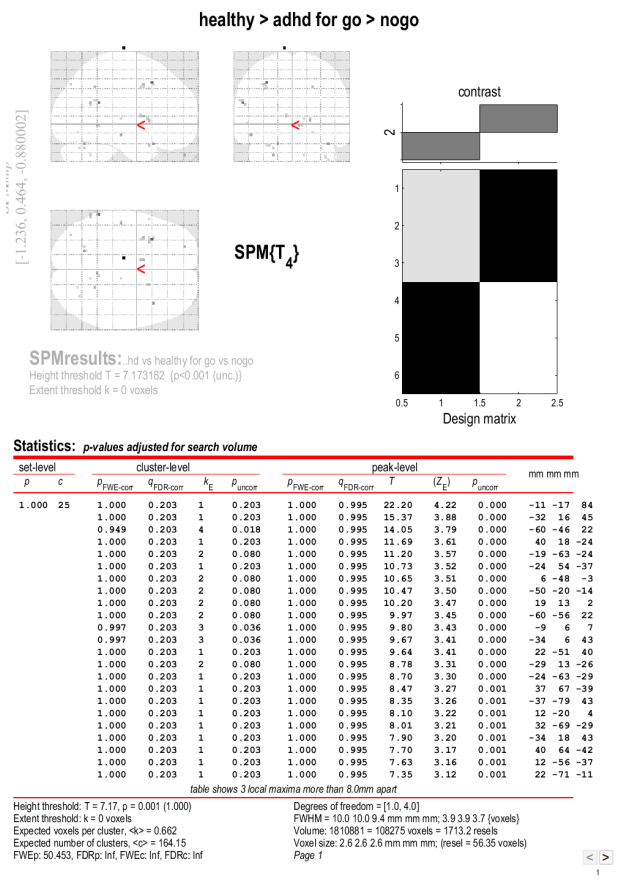
Consistency with task expectations

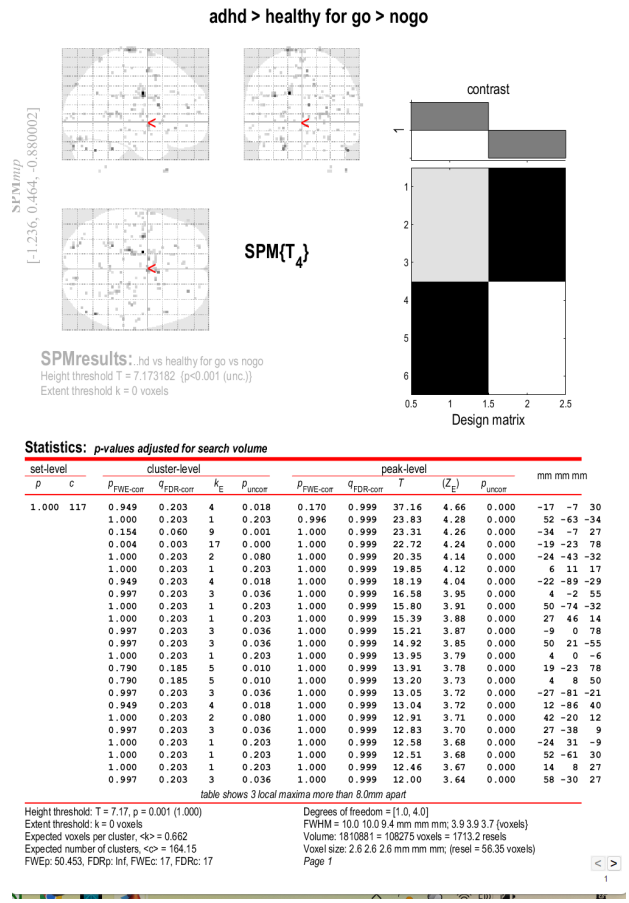
- Uncorrected activation for go > nogo aligns with known engagement of fronto–parietal–motor control circuits in Go/No-go paradigms, though canonical inhibitory contrasts typically emphasize nogo > go in right-lateralized control regions.[3]
- The observed left-hemisphere peaks may reflect task-specific motor demands and response selection processes that are commonly reported in Go/No-go literature.

Two-sample t tests

In this study contrasting ADHD and controls during inhibitory tasks, meta-analyses report reduced activation in right inferior frontal cortex and supplementary motor areas for ADHD, which defines a principled a priori hypothesis space for targeted second-level contrasts.[9]

- Applying SPM's two-sample design to subject contrast images enables voxelwise tests of ADHD < Control and the reverse, with correction for multiple comparisons to quantify group effects aligned with prior literature.





Neither direction of the two-sample t test (subjects 1–3 vs. 4–6) shows any whole-brain significant group differences after multiple-comparisons correction; both "ADHD > healthy" and "healthy > ADHD" maps display FWE- and FDR-corrected p values ≈ 1.000 at cluster and peak levels.

Model and thresholds

- The design matrix indicates a standard two-sample second-level model with df = 4 ($n1 + n2 - 2 = 6 - 2$) and independent groups, matching SPM's recommended setup for between-subject comparisons.
- Both results are thresholded at $p < 0.001$ uncorrected ($T \approx 7.17$ for df = 4) with an extent threshold k = 0, smoothness around FWHM ≈ 10 × 10 × 9.4 mm, and a search volume of ~180,881 voxels (~56 resels), which together impose a stringent multiple-comparisons burden.

Uncorrected findings

- In the "ADHD > healthy" contrast, numerous tiny clusters exceed the uncorrected height threshold (e.g., peaks near x,y,z ≈ −17, −7, 30; 52, −63, −34; and −19, −23, 78), but none survive cluster-level or peak-level FWE/FDR correction, so these foci are exploratory only.
- In the "healthy > ADHD" contrast, a similar pattern appears with scattered suprathreshold voxels (e.g., peaks near −11, −17, 84; −32, 16, 45; and −60, −46, −22) that do not pass whole-brain correction, indicating no reliable group difference at the chosen thresholds.
- With only three subjects per group (df = 4), the analysis is severely underpowered for whole-brain voxelwise inference, making the absence of corrected findings unsurprising despite high uncorrected T values.
- Given Go/No-go literature, any hypothesized group differences would most plausibly emerge in inhibitory control regions (e.g., right inferior frontal cortex, pre-SMA), but detecting them typically requires larger samples or ROI-level tests.

Interpretations

Observed subject-level variability alongside expected group-level patterns reflects the balance between individual noise and the reproducible engagement of control networks during response inhibition, which is why second-level summary statistics are standard in SPM.[7]

Group differences in ADHD research frequently localize to right-lateralized frontal and medial motor control regions, consistent with meta-analytic evidence of hypoactivation during inhibition, while mixed results in adults highlight the influence of sample size, medication, and task parameters.[6]

Limitations

Single-subject contrast maps cannot identify clinical status, and diagnostic labeling requires validated behavioral and clinical assessments beyond imaging, reinforcing the need for group or predictive-model frameworks.[10]

The N=6 sample severely limits power for whole-brain voxelwise inference, and the absence of demographic/clinical covariates precludes modeling factors such as age, sex, or medication status that influence inhibitory-control activation. And heterogeneity in ADHD phenotypes and medication status can attenuate or shift effects across frontal and subcortical regions.

Conclusion

This study followed standard SPM workflows to preprocess data, model condition onsets, generate subject-level contrasts for go and nogo events, and conduct second-level one-sample and illustrative two-sample tests appropriate for population inference in inhibitory control research.

Results and interpretations are consistent with established Go/No-go literature highlighting pre-SMA and inferior frontal contributions to response inhibition and meta-analytic evidence of reduced activation in these regions in ADHD, while emphasizing careful design reporting, multiple-comparisons control, and caution against single-subject diagnostic claims. The addition of a clinical vignette and comprehensive sample reporting aligns the study with both the assignment's narrative requirement and best-practice guidelines for reproducible neuroimaging.